Prob. 2 (2aT, Ch 5, sect. 7) (change of var-s.

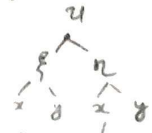
Given $u(x,y)$, we want to go to $u(\xi,\eta)$ given

Sol-n: $\xi = \xi(x,y)$, $\eta = \eta(x,y)$.

(a) Write up, in your own words, the derivation of

(1) $$u_{yy} = u_{\xi\xi}(\xi_y)^2 + u_{\xi\eta}(2\xi_y\eta_y) + u_{\eta\eta}(\eta_y)^2 + \dots .$$

Note that for $u = u(\xi(x,y), \eta(x,y))$:



$u_y = u_\xi \cdot \xi_y + u_\eta \cdot \eta_y$ (basically by chain rule & total differentiation)

So

Product rule

$$u_{yy} = \frac{\partial}{\partial y}[u_y] = \frac{\partial}{\partial y}[u_\xi \cdot \xi_y + u_\eta \cdot \eta_y] = \frac{\partial}{\partial y}[u_\xi \cdot \xi_y] \oplus \frac{\partial}{\partial y}[u_\eta \cdot \eta_y] =$$

$$= \frac{\partial}{\partial y}[u_\xi] \cdot \xi_y + u_\xi \frac{\partial}{\partial y}[\xi_y] \oplus \frac{\partial}{\partial y}[u_\eta] \cdot \eta_y + u_\eta \cdot \frac{\partial}{\partial y}[\eta_y] =$$

$$[u_{\xi\xi} \cdot \xi_y + u_{\xi\eta} \cdot \eta_y] \cdot \xi_y + u_\xi \cdot \xi_{yy} \oplus [u_{\eta\xi} \cdot \xi_y + u_{\eta\eta} \cdot \eta_y] \cdot \eta_y + u_\eta \cdot \eta_{yy}$$

$$= u_{\xi\xi} \cdot \xi_y^2 + u_{\xi\eta}\xi_y\eta_y + u_\xi \cdot \xi_{yy} \oplus u_{\eta\xi} \cdot \xi_y \cdot \eta_y + u_{\eta\eta} \cdot \eta_y^2 + u_\eta \cdot \eta_{yy} =$$

$$= u_{\xi\xi}\xi_y^2 + u_{\xi\eta}(\xi_y\eta_y + \xi_y \cdot \eta_y) + u_{\eta\eta} \cdot \eta_y^2 + u_\xi \xi_{yy} + u_\eta \eta_{yy} =$$

(2) $$= u_{\xi\xi}\xi_y^2 + u_{\xi\eta}(2 \cdot \xi_y\eta_y) + u_{\eta\eta}\eta_y^2 + u_\xi\xi_{yy} + u_\eta\eta_{yy} .$$

$\equiv (1)$ ✓. ▨

(b) Now use (similar result of part (a))

(3) $$u_{xx} = u_{\xi\xi}\xi_x^2 + u_{\xi\eta}(2\xi_x\eta_x) + u_{\eta\eta}\eta_x^2 + u_\xi\xi_{xx} + u_\eta\eta_{xx}$$

(4) $$u_{xy} = u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) + u_{\eta\eta}\eta_x\eta_y + u_\xi\xi_{xy} + u_\eta\eta_{xy}$$

(5) $$u_{yy} = u_{\xi\xi}\xi_y^2 + u_{\xi\eta}(2\xi_y\eta_y) + u_{\eta\eta}\eta_y^2 + u_\xi\xi_{yy} + u_\eta\eta_{yy}$$

to have the canonical form for

(6) $$a u_{xx} + b u_{xy} + c u_{yy} = A u_{\xi\xi} + B u_{\xi\eta} + C u_{\eta\eta}$$

where

(7) $$\begin{cases} A = a\xi_x^2 + b\xi_x\xi_y + c\xi_y^2 \\ B = 2a\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + 2c\xi_y\eta_y \\ C = a\eta_x^2 + b\eta_x\eta_y + c\eta_y^2 \end{cases}$$

Sol-n. If we replace (3), (4), (5) into the LHS of eq-n (6):

$$a(u_{\xi\xi}\xi_x^2 + 2u_{\xi\eta}\xi_x\eta_x + u_{\eta\eta}\eta_x^2) + b(u_{\xi\xi}\xi_x\xi_y + u_{\xi\eta}(\xi_x\eta_y + \xi_y\eta_x) +$$

$$+ u_{\eta\eta}\eta_x\eta_y) + c(u_{\xi\xi}\xi_y^2 + 2u_{\xi\eta}\xi_y\eta_y + u_{\eta\eta}\eta_y^2) =$$

$$= \overbrace{(a\xi_x^2 + b\xi_x\xi_y + c\xi_y^2)}^{A}\cdot u_{\xi\xi} + \overbrace{(a\eta_x^2 + b\eta_x\eta_y + c\eta_y^2)}^{C}\cdot u_{\eta\eta} +$$

$$+ \underbrace{(2a\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + 2c\xi_y\cdot\eta_y)}_{B}\cdot u_{\xi\eta} =$$

$= A u_{\xi\xi} + B u_{\xi\eta} + C u_{\eta\eta}$ where A, B, C are the same expressions given in (7). ∎

(c) Let $\Delta' = B^2 - 4AC$ where A, B, C have been given in (7), p(b).

(8) & $\Delta = b^2 - 4ac$ where a, b, c have been given in (6), part (b).

Show that

$$\boxed{\Delta' = \Delta\cdot J^2} \quad \text{or} \quad B^2 - 4AC = (b^2 - 4ac)\cdot(\xi_x\eta_y - \xi_y\cdot\eta_x)^2.$$

Why is the result so important?

Sol-n: Let me directly compute $B^2 - 4AC$ using (7):

So by (7)

(9) $$B^2 = \left(2a\,\xi_x\eta_x + b(\xi_x\eta_y + \xi_y\eta_x) + 2c\,\xi_y\eta_y\right)^2$$

(10) $$4AC = \left(4a\,\xi_x^2 + 4b\,\xi_x\xi_y + 4c\,\xi_y^2\right)\cdot\left(a\,\eta_x^2 + b\,\eta_x\eta_y + c\,\eta_y^2\right)$$

(9') $$B^2 = 4a^2(\xi_x\eta_x)^2 + b^2(\xi_x\eta_y + \xi_y\eta_x)^2 + 4c^2(\xi_y\eta_y)^2 +$$

$$+ 2\cdot(2a\,\xi_x\eta_x)\cdot\left(b(\xi_x\eta_y + \xi_y\eta_x)\right) + 2\left(b(\xi_x\eta_y + \xi_y\eta_x)\right)\cdot(2c\,\xi_y\eta_y) +$$

$$+ 2\cdot(2a\,\xi_x\eta_x)\cdot(2c\,\xi_y\eta_y) \boxed{=} \underline{4a^2(\xi_x\eta_x)^2} + \underline{\underline{b^2(\xi_x\eta_y + \xi_y\eta_x)^2}} + \underline{4c^2(\xi_y\eta_y)^2}$$

$$+ \underline{4ab\left(\xi_x^2\cdot(\eta_x\eta_y) + (\xi_x\xi_y)\cdot\eta_x^2\right)} + 4bc\left((\xi_x\xi_y)\cdot\eta_y^2 + \xi_y^2(\eta_x\eta_y)\right) +$$

$$+ \underline{8ac\,(\xi_x\eta_x)\cdot(\xi_y\eta_y)}.$$

(10') $$4AC = \underline{4a^2(\xi_x\eta_x)^2} + \underline{4ab\,\xi_x^2(\eta_x\eta_y)} + \underline{4ac\,(\xi_x\eta_y)^2} +$$

$$+ \underline{4ab\,\eta_x^2(\xi_x\xi_y)} + \underline{\underline{4b^2(\xi_x\xi_y\cdot\eta_x\eta_y)}} + 4bc\,\eta_y^2\cdot(\xi_x\xi_y) +$$

$$+ \underline{4ac\,(\xi_y\eta_x)^2} + 4bc\,\xi_y^2\cdot(\eta_x\eta_y) + \underline{4c^2(\xi_y\eta_y)^2} =$$

$$\boxed{=} \underline{4a^2(\xi_x\eta_x)^2} + \underline{\underline{4b^2(\xi_x\xi_y\cdot\eta_x\eta_y)}} + \underline{4c^2(\xi_y\eta_y)^2} +$$

$$+ \underline{4ab\left(\xi_x^2(\eta_x\eta_y) + \eta_x^2(\xi_x\xi_y)\right)} + \underline{4ac\left((\xi_x\eta_y)^2 + (\xi_y\eta_x)^2\right)}$$

$$+ 4bc\left(\xi_y^2(\eta_x\eta_y) + \eta_y^2(\xi_x\xi_y)\right)$$

So, (9') − (10') $= B^2 - 4AC$ gives us

$$\underline{\underline{b^2\left[(\xi_x\eta_y+\xi_y\eta_x)^2-4(\xi_x\xi_y\cdot\eta_x\eta_y)\right]}}\ +$$

$$+\ 4ac\left\{2(\xi_x\eta_x)\cdot(\xi_y\cdot\eta_y)-(\xi_x\eta_y)^2+(\xi_y\eta_x)^2\right\}$$

$$=b^2\left[(\xi_x\eta_y)^2+2(\xi_x\eta_y)\cdot(\xi_y\cdot\eta_x)+(\xi_y\eta_x)^2-4\cdot(\xi_x\eta_y)\cdot(\xi_y\eta_x)\right]-$$

$$-4ac\left\{(\xi_x\eta_y)^2-2(\xi_x\eta_y)\cdot(\xi_y\eta_x)+(\xi_y\eta_x)^2\right\}=$$

$$=b^2\underbrace{\left[(\xi_x\eta_y)-(\xi_y\eta_x)\right]^2}_{J^2}-4ac\left\{(\xi_x\eta_y-\xi_y\eta_x)^2\right\}=$$

$$=b^2\overbrace{\begin{vmatrix}\xi_x & \xi_y\\ \eta_x & \eta_y\end{vmatrix}^2}^{J^2}-4ac\overbrace{\begin{vmatrix}\xi_x & \xi_y\\ \eta_x & \eta_y\end{vmatrix}^2}^{J^2}=(b^2-4ac)\cdot J^2.$$

So

$$\Delta'=B^2-4AC=(b^2-4ac)J^2=\Delta\cdot J^2\ \Rightarrow$$

$$\Delta'=\Delta\cdot J^2,\quad \text{as desired.}\quad \blacksquare$$